Three-dimensional data of wire-cut surface scans under the confocal microscope

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Abstract

Wire-cut striation marks provide evidence for forensic source identification, but publicly available datasets and reproducible quantitative workflows for analyzing these marks are limited. We present a dataset of 134 three-dimensional topographic scans in `x3p` format from aluminum wires cut with five KWS-105 wire cutters. The main experiment contains 120 scans: two replicates at each of three cutting locations for each of the 20 tool-edge combinations. Fourteen additional scans assess staging and acquisition variability for a randomly selected tool and location. We provide additional scan metadata, manually extracted profiles, processed signals, cross-correlation function (CCF) values for pairwise aligned signals, images, and analysis code. The accompanying workflow covers scan inspection, profile and signal extraction, alignment, pairwise similarity visualization, CCF distributions for same-source and different-source comparisons, and variability assessment across replicates, stagings, and acquisitions. The dataset and workflow support the development, evaluation, and reproducible comparison of quantitative methods for wire-cut evidence.

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Background & Summary

In forensic analysis, marks left at crime scenes play an important role, and forensic examiners are particularly interested in identifying their origin, a process known as the Source Identification Problem in Forensic Science[1]. The forensic science community focuses on two distinct problems: the **specific source problem**, in which examiners aim to determine whether a mark was left by a particular tool, and the **common source problem**, in which examiners aim to determine whether two marks were left by the same tool. Currently, accepted practice for both problems relies on visual inspection of physical items under a comparison microscope. Examiners then report their conclusions according to the Theory of Identification[2], developed by the Association of Firearm and Toolmark Examiners (AFTE), using one of three categories: *identification*, in which the marks are believed to have been made by the same source; *elimination*, in which the marks are believed to have been made by different tools; and *inconclusive*, in which the similarities or dissimilarities between marks are insufficient to support either identification or elimination. This process has been criticized by the National Research Council (NRC)[3] and the President's Council of Advisors on Science and Technology (PCAST)[4] for its lack of objectivity and absence of quantifiable measurements, such as error rates.

In discussing error rates, Biedermann[5] differentiates between internal and external perspectives. The external perspective allows general statements based on black-box studies that relate examiners' conclusions to ground truth; however, it does not consider the evidence in a particular case. Introducing an internal perspective requires evaluating similarity between pairs of specimens with known ground truth. Algorithms allow these similarities to be quantified objectively and reproducibly.

Advances in microscopy have made it possible to move from optical two-dimensional imaging to digital three-dimensional topographic scanning, allowing algorithmic assessment of the similarity between pairs of forensic pattern evidence[6]. This transition has already supported practical advances in bullet and cartridge comparisons, including visual inspection of bullet line scans from three-dimensional topographic images[7], the use of features extracted by fast Fourier transform (FFT) as the basis of an artificial intelligence system[8], an automated pilot study based on consecutively matching striae[9] (a feature also used by some laboratories in casework), and the incorporation of three-dimensional imaging into routine casework at the Alabama Department of Forensic Sciences (ADFS)[10].

When cut wires are found at a crime scene, understanding the characteristics of marks left by bladed tools becomes essential. When a blade cuts a wire, it leaves striations on the surface, as shown in Figure 1. These striations provide a basis for similarity assessment and source identification. Similar scans have supported the development of quantitative methods for analyzing firearms and other toolmark evidence. For example, studies of screwdriver striations have examined key factors affecting comparisons, including angles of attack[11], rotational axes[12], and cutting directions[13]. Foundational research on forensic toolmarks has also emphasized data collection and analysis, including the collection and distribution of datasets for bullet and toolmark analysis[14,15], numerical methods for improving accuracy and consistency[16,17], and techniques for quantifying similarity[18,19].

Data collected at crime scenes lack the known ground truth needed to estimate error rates; therefore, controlled studies are needed. In this study, we provide a publicly available dataset of wire-cut scans together with a systematic quantitative processing and comparison workflow. The shared materials include raw scans, metadata, manually extracted profiles, processed signals, pairwise cross-correlation function (CCF) values, derived images, and analysis code.

The Methods section describes sample collection, scanning, profile extraction, signal processing, and CCF-based alignment. Data Records describes the shared files and repository organization. Data Overview presents descriptive summaries of pairwise CCF values. Technical Validation assesses signal-processing and alignment behavior and variability across replicates, stagings,

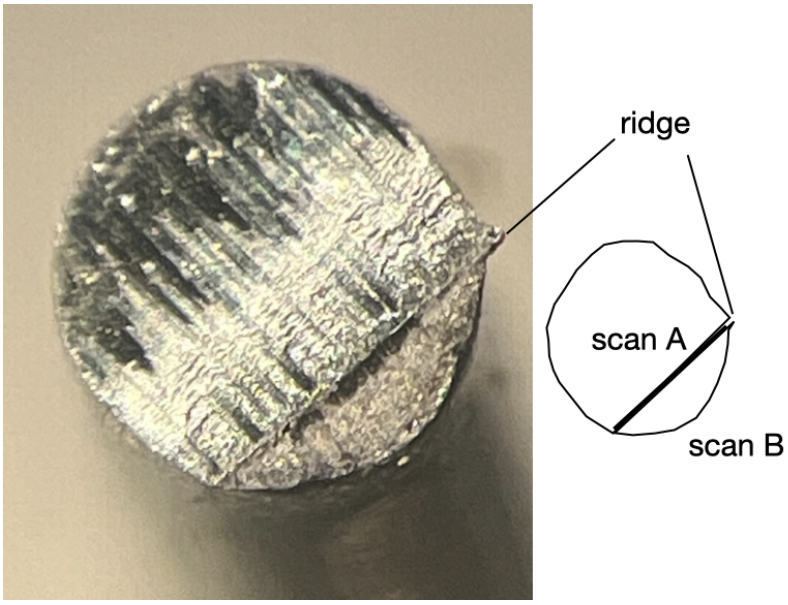


Figure 1. Microscopic wire-cut striations. Close-up of striations left by a blade on the cut end of a wire; the schematic identifies the ridge separating the two roof-like surfaces scanned separately as scans A and B.

and acquisitions. Usage Notes provides practical guidance for working with the shared files, and Code Availability identifies the custom processing and validation code. The dataset and workflow may support future quantitative research on wire-cut evidence.

Data Acquisition and Processing

Five Kaiweets KWS-105 multipurpose tools with 0.5-inch carbon-steel cutting blades were used to cut 16-gauge anodized pure-aluminum wire.

While aluminum is not commonly used for electric wiring because of its poor conductivity, it was selected for its nontoxicity and suitability for receiving cutting marks and retaining them during subsequent handling. Pure aluminum has a Brinell hardness of approximately 15 HB, which places it between lead (approximately 5 HB) and copper (approximately 35 HB).

Scan acquisition

The blades of each of the five wire cutters were labeled A, B, C, and D (see Figure 2 (a)). B opposes A; the reverse sides are labeled D for A and C for B. Three locations on each blade were marked (inner, middle, and outer), as shown in Figure 2 (a). Each four-inch wire segment was cut in half, creating two cut sides, AB and CD. The cut surface of the wire is not flat; under the microscope, it reveals a roof-like structure with a ridge separating the markings made by each blade (see Figure 1). The two roof sides of each cut surface were scanned separately, yielding scans A and B from cut side AB and scans C and D from cut side CD. As an example, the four scans resulting from cut 1 by tool 1 are shown in Figure 2 (c).

The cut surfaces were scanned at 20× magnification using a Sensofar confocal microscope, shown in Figure 2 (b). The scans were exported in `x3p` format (ISO 25178-72:2017/Amd 1:2020) with a lateral resolution of 0.645 × 0.645 micrometers per pixel and a vertical resolution of 0.1 micrometer.

The main experiment therefore consists of 5 tools × 3 locations × 2 replicates × 4 cut surfaces = 120 scans from 30 cuts. Tool 2 and the middle cutting location were randomly selected for variability assessment. Fourteen additional scans of the second replicate at this tool and

location were collected across repeated stagings and acquisitions, giving 134 scans in total. These repeated scans do not represent additional cuts; their acquisition design is described in Technical Validation later.

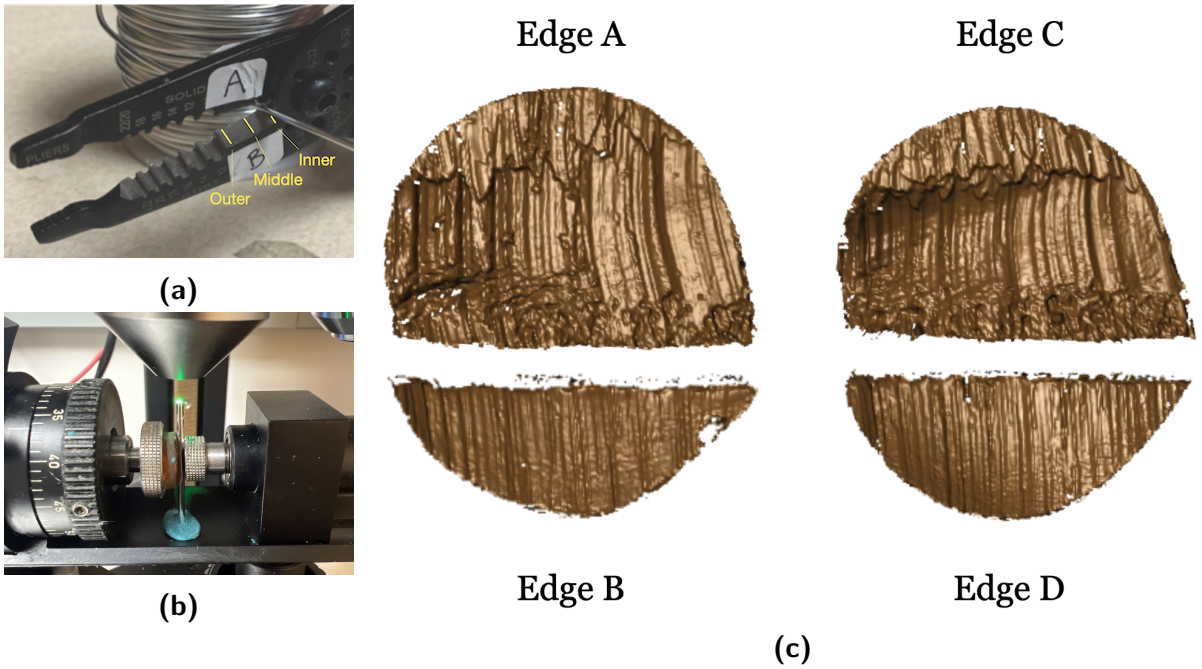


Figure 2. Wire-cut sampling and scanning workflow. (a) A Kaiweets KWS-105 wire cutter with inner, middle, and outer locations marked. (b) A confocal microscope used to scan wire-cut surfaces. (c) Four scans of cutting marks labeled according to the corresponding cutting edges A, B, C, and D.

Each scan identifier specifies the tool, edge, cutting location, replicate, and, where applicable, staging and acquisition according to the following convention:

```
T__W-L_-R_[-S_[-A_]]
 ||   |  |   |   |
 ||   |  |   |   +-- Acquisition: 2, 3, or 4
 ||   |  |   +------ Staging: 2 or 3
 ||   |  +---------- Replicate: 1 or 2
 ||   +------------- Location: I (inner), M (middle), or O (outer)
 |+----------------- Edge: A, B, C, or D
 +------------------ Tool: 1, 2, 3, 4, or 5
```

Square brackets denote optional suffixes and are not part of the identifiers. Staging and acquisition suffixes are omitted when their corresponding value is 1; these values are recorded explicitly in the metadata and signal files. In our scans, acquisition suffixes occur only at staging 3. For example, T1AW-LI-R1 denotes the first replicate of a piece cut with edge A of tool 1 at the inner location, at staging 1 and acquisition 1. T2AW-LM-R2-S3-A2 denotes the second acquisition at the third staging of replicate 2, cut with edge A of tool 2 at the middle location.
To minimize errors during data collection, two team members cut and labeled the wire pieces together. One team member scanned the wire cuts and assigned names to the scans as described.

Signal extraction

Signals were extracted from each surface scan using a two-step procedure. First, a representative profile was identified along a line across the surface perpendicular to the dominant striation direction. This line was chosen manually to avoid scanning artifacts (e.g., holes) and cutting

artifacts (e.g., rips caused by rolling the wire). Along this line, surface values were assigned at equally spaced positions using the observed surface value at the nearest position on the surface grid.

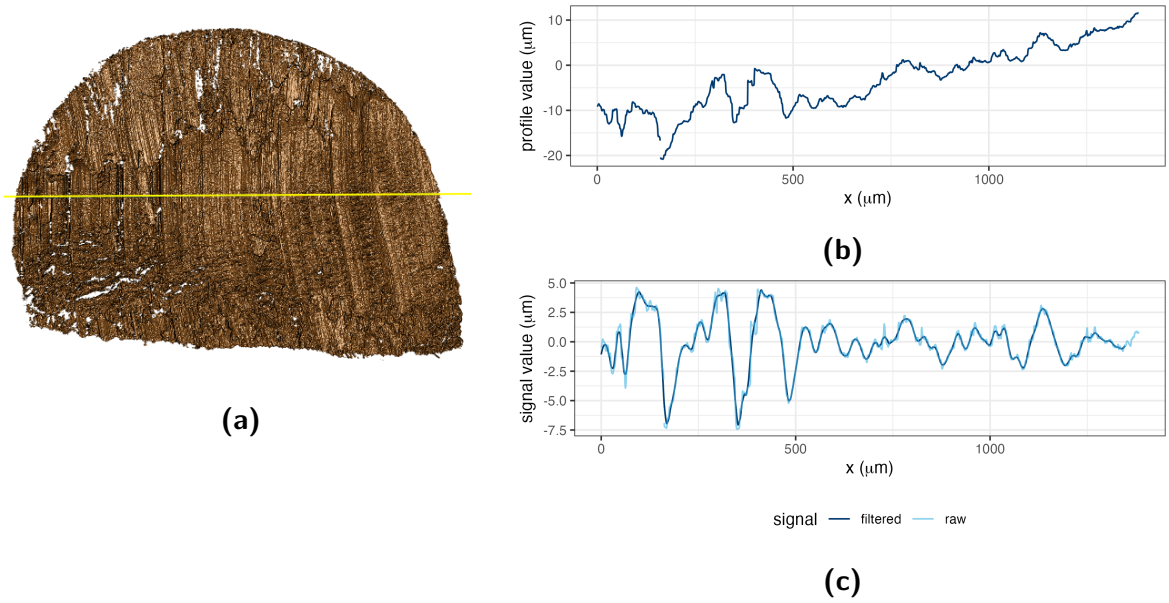


Figure 3. Profile and signal extraction. (a) A yellow profile line drawn across the scan striations. (b) The profile function extracted along the line in (a). (c) Raw and filtered signals obtained from the profile function in (b).

In a second step, we extract the scan's signal by removing trends and outliers from the surface values of the profile with the help of two LOWESS filters with bandwidths of $400\mu m$ and $40\mu m$, respectively[20]. All smoothers are susceptible to boundary issues, particularly when outliers are present. Steep changes at the end of a signal have a high potential to attract another signal during alignment, causing misalignments. To avoid potential artificial "hooks'' at either end of a signal, two-sample t-tests were applied to the surface values in the outermost 2.5% intervals along the horizontal axis. Specifically, the summary statistics (sample size, mean, and standard deviation) of the signal values within the intervals $[0, 2.5)\%$, $[2.5, 5)\%$, $(95, 97.5]\%$, and $(97.5, 100]\%$ were calculated. At each end, the test compared the outermost 2.5% interval with the adjacent interval. The signal values in the outermost interval were retained only when the corresponding p-value was above 0.01, indicating no significant shift.
The widths of these intervals correspond to the choice of bandwidths and the length of the signal. The interval widths were determined using cross-validation.
The resulting signals are used for subsequent comparisons with other signals.
Figure 3 shows a scan on the left (a) with the location of the manually chosen profile line in yellow. The corresponding profile and signal are plotted in (b) and (c).
One profile for each scan is included in folder `profiles/` as a csv file in the repository using the same naming convention as the scans, with the precursor of `profiles-`. More details on the exact format of these files are provided below.
A single file containing signals for all 134 scans is available in `data-derived/wire-signals.csv.zip`.

Pairwise alignment of signals

Before signals from different scans can be compared, they need to be aligned. For each pair of signals, the value of the cross-correlation function (CCF) is calculated across relative shifts. The shift corresponding to the maximum CCF is retained as the alignment position, while the

maximized cross-correlation function, CCF_{max}, gives a measure of the similarity of the signals, with a value of 1 indicating complete similarity (apart from linear scaling) and a value of 0 indicating (statistical) independence.

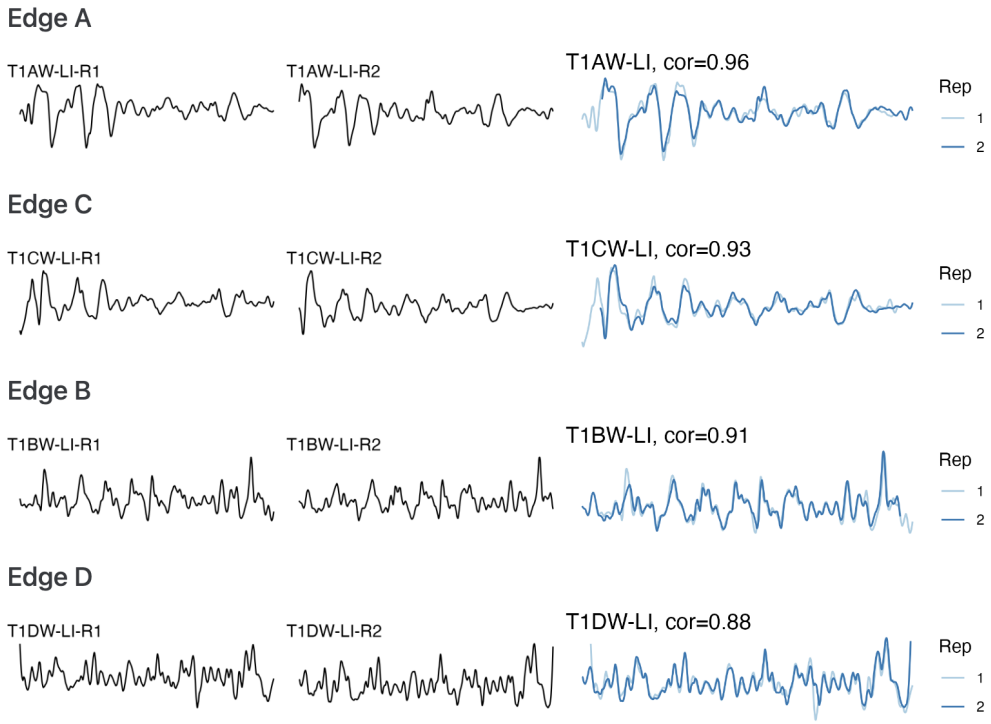


Figure 4. Same-source signal alignment. Rows correspond to cut surfaces A, C, B, and D. The first two columns show signals extracted from scans corresponding to the replicate cuts of tool 1 at location I; the third column overlays the aligned signals and reports their CCF values.

Figure 4 shows all signals corresponding to cuts 1 and 2 made with tool 1 at the location closest to the pivot area (inner location). Each of the four edges corresponds to one row in the figure. The lines in the first two columns correspond to the extracted signals for cuts 1 and 2, respectively. The third column shows the two signals aligned at the location of the maximal CCF. The cross-correlation value for each pair is included as part of the title of the chart.
The file `data-derived/wire_pairwise_ccf.csv` contains the CCF similarity scores of all pairwise aligned signals.

Data Records

The repository record on Iowa State University DataShare[21] at `https://doi.org/10.25380/iastate.34025028` contains the `x3p` scans, metadata, extracted profiles, processed signals, pairwise CCF values, derived images, analysis code, and supporting documentation and environment files. The data files, formats, and folder organization are summarized in Table 1. All 134 scans share the root-level `scans/`, `profiles/`, `pngs/`, and `profile-images/` folders, a single `meta.csv`, and the derived files in `data-derived/`. The same file structures describe both the main experiment and the repeated scans used for variability assessment. The 120 main-experiment scans have `staging = 1` and `acquisition = 1`; the 14 additional scans have staging 2 or 3. Scan identifiers link the metadata, profiles, signals, and images, and both identifier columns in the pairwise file refer to this shared set.
The dataset is licensed under the Creative Commons Attribution 4.0 International (CC BY 4.0) license.

Table 1. Structure of available data and files.

Data type	Description	Section
Raw data		
`scans/`	134 topographic 3D scans from 30 aluminum wire cuts, including 14 repeat scans (`x3p` format)	Scan acquisition
`meta.csv`	scan identifiers, paths, tool, edge, location, replicate, staging, acquisition, and pixel dimensions (CSV format)	Scan acquisition
Manual derivatives		
`profiles/`	134 manually extracted profiles, one per scan (CSV format)	Signal extraction
Computational derivatives in the folder 'data-derived/'		
`wire-signals.csv.zip`	profile values and derived signals for all 134 scans (compressed CSV format)	Signal extraction
`wire_pairwise_ccf.csv`	17,956 ordered pairwise CCF values for all 134 signals (CSV format)	Pairwise alignment of signals
Image files		
`pngs/`	134 images of 3D wire-cut scans (PNG format)	Scan acquisition
`profile-images/`	134 scan images showing the manually selected profile lines (PNG format)	Signal extraction

The `meta.csv` file contains one record for each of the 134 scans, with ten variables described in Table 2.

Table 2. Structure of `meta.csv`.

Variable	Type	Description
`ID`	text	Unique scan identifier defined by the naming convention in Scan acquisition
`x3p_path`	text	Path to the `x3p` scan relative to the data repository root
`tool`	integer	Cutting tool, one of 1 through 5
`edge`	character	Cutting edge, one of A, B, C, or D
`location`	character	Cutting location: I (inner), M (middle), or O (outer)
`replicate`	integer	Replicate cut, either 1 or 2
`staging`	integer	Staging of the same cut surface under the microscope: 1, 2, or 3; 1 is omitted from the scan identifier
`acquisition`	integer	Acquisition within a staging: 1 through 4; 1 is omitted from the scan identifier
`height`	integer	Height of the scan in pixels
`width`	integer	Width of the scan in pixels

The `profiles/` folder contains 134 CSV files, one manually extracted profile per scan, with one sampled point per row and 266,394 rows in total. Their four variables are described in Table 3

in CSV column order.

Table 3. Structure of the `profiles-*.csv` files.

Variable	Type	Description
`ID`	text	Source scan identifier, matching `ID` in `meta.csv`; constant within each file
`x`	numeric	Horizontal coordinate along the profile in micrometers, starting at zero in increments of 0.645
`y`	numeric	`y` coordinate, constant and set to zero
`value`	numeric	Observed surface height in micrometers; `NA` denotes a missing value

The `data-derived/wire-signals.csv.zip` archive contains `data-derived/wire-signals.csv`, with 266,394 rows containing profile values and signals for all 134 scans. Table 4 describes its 13 variables in CSV column order.

Table 4. Structure of the signal CSV in `wire-signals.csv.zip`.

Variable	Type	Description
`ID`	text	Source scan identifier, matching `ID` in `meta.csv` and the profile files
`tool`	integer	Cutting tool, one of 1 through 5
`edge`	character	Cutting edge, one of A, B, C, or D
`location`	text	Cutting location: I (inner), M (middle), or O (outer)
`replicate`	integer	Replicate cut, either 1 or 2
`staging`	integer	Staging of the cut surface: 1, 2, or 3, matching `meta.csv`
`acquisition`	integer	Acquisition within a staging: 1 through 4, matching `meta.csv`
`x`	numeric	Horizontal coordinate along the profile in micrometers, starting at zero in increments of 0.645
`y`	numeric	`y` coordinate, constant and set to zero
`value`	numeric	Observed surface height in micrometers
`detrend`	numeric	Surface height after subtracting the fitted linear trend in `value` versus `x`, in micrometers
`raw_sig`	numeric	Residual in micrometers after removing the 400-micrometer LOWESS trend from `detrend`, before final smoothing and boundary exclusion
`sig`	numeric	Signal after 40-micrometer smoothing of `raw_sig` and boundary exclusion, in micrometers

The `data-derived/wire_pairwise_ccf.csv` file contains all 17,956 ordered pairs of the 134 signals, including self-comparisons and both orders of each distinct pair. The main-experiment subset contains 14,400 ordered pairs of the 120 signals at staging 1 and acquisition 1. Table 5 describes the file's 12 variables in CSV column order.

Table 5. Structure of the `wire_pairwise_ccf.csv` file.

Variable	Type	Description
`ID1`	text	Scan identifier for the first signal, matching `ID` in `meta.csv`
`ID2`	text	Scan identifier for the second signal, matching `ID` in `meta.csv`
`ccf`	numeric	Maximized cross-correlation function value for the aligned signals
`lag`	integer	Relative shift at which `ccf` is maximized, in profile-sample increments
`same_tool`	logical	Whether both signals were produced by the same tool
`same_edge`	logical	Whether both signals were produced by the same tool and edge
`same_location`	logical	Whether both signals were produced by the same tool, edge, and location
`same_replicate`	logical	Whether `same_location` is true and the replicate cut is the same
`same_staging`	logical	Whether `same_replicate` is true and the staging is the same
`same_acquisition`	logical	Whether `same_staging` is true and the acquisition is the same
`same_source`	logical	Whether `same_location` is true; replicate, staging, and acquisition may differ
`edge_type`	text	Edge-length pairing: Long, Short, or Mixed

Data Overview

As a descriptive summary of the 120 main-experiment scans at staging 1 and acquisition 1, Figure 5 displays the CCF distributions for same-source and different-source comparisons. The density summary counts each distinct unordered pair once and excludes self-comparisons.

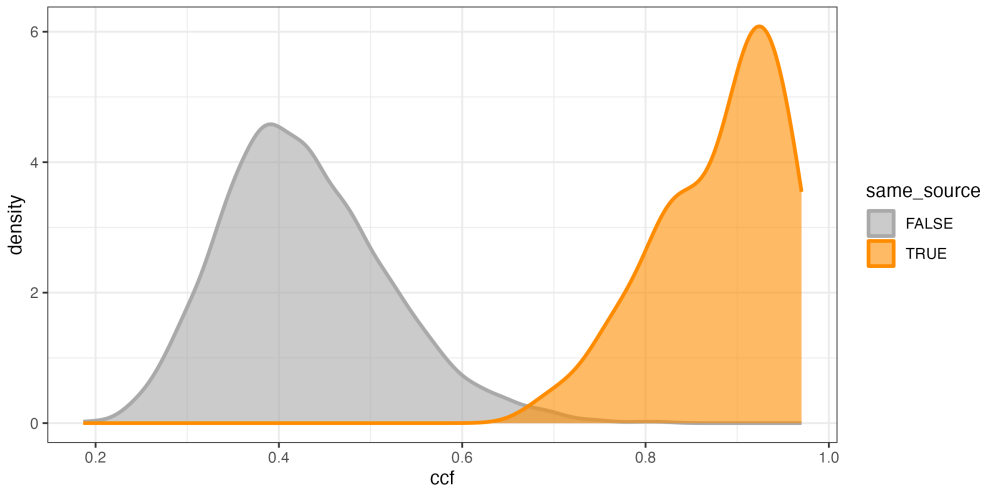


Figure 5. CCF density distributions. Comparisons use the 120 main-experiment scans at staging 1 and acquisition 1. Kernel density estimates are shown for different-source comparisons in gray and same-source comparisons in orange. Each distinct unordered pair is counted once, excluding self-comparisons.

Technical Validation

Technical validation was conducted in two areas: (1) signal processing and alignment and (2) variability assessment.

Scan processing and signal alignment were validated using the 120 main-experiment scans at staging 1 and acquisition 1 as follows. High CCF values were expected between signals from the same source and low values between signals from different sources. Figure 4, shown earlier in Pairwise alignment of signals, meets the expected pattern for same-source signals. As an example of a different-source comparison, T1AW-LI-R1 and T2AW-LI-R1 were selected. Figure 6 (a) and (b) show the selected scans. Their signal alignment, shown in Figure 6 (c), has a CCF value of 0.2, which is as low as expected. For the following summaries, edges A and C were categorized as long, edges B and D as short, and comparisons pairing one long edge with one short edge as mixed. The resulting CCFs for all pairwise comparisons were also summarized using boxplots and receiver operating characteristic (ROC) curves. These summaries count each distinct unordered pair once and exclude self-comparisons. Figure 7 groups CCF distributions from left to right by same source, different location, different edge, and different tool, in order of decreasing expected similarity, with the paired edges identified on the horizontal axis. Comparisons between different edges remain difficult to interpret, even when CCF values are high, because edges A and C, or B and D, of some tools may produce similar striations that are difficult to distinguish. Figure 8 shows classification performance separately for long and short edge classes, comparing same-source pairs with different-tool pairs. The ROC curves plot sensitivity, or the true positive rate, against the false positive rate (FPR; 1 - specificity). The curves are close to the upper-left corner, indicating strong classification performance. For the combined comparison set, a threshold of 0.589 controls the FPR below 0.05 with a false negative rate (FNR) of 0, whereas a threshold of 0.658 controls the FPR below 0.01 with an FNR of 0.02. These results are consistent with the expected pattern and validate the pipeline for processing and aligning signals.

The second validation assessed variability in the pipeline across replicates, stagings, and acquisitions. Tool 2 and the middle location were the randomly selected tool and cutting location for this assessment. The 22 scans from tool 2 at the middle location were selected from the shared files for this assessment: eight main-experiment scans and 14 additional scans of replicate 2. Replicate variability was assessed using two different cuts at the same edge and position. The comparison in Figure 4 in Pairwise alignment of signals illustrates this assessment. To assess staging variability, replicate 2 was rescanned under the confocal microscope; the staging label S identifies these repeated stagings in the naming convention. The S1 designation is omitted from labels, so R2 is equivalent to R2-S1. For all four edges of tool 2 at the middle location, three stagings (R2, R2-S2, and R2-S3) were created. Their pairwise CCFs were summarized by the mean and standard deviation (SD) using the comparison categories defined below.

Within each staging, the effect of different acquisitions was further examined by keeping the samples on the confocal microscope while varying the lighting conditions. The suffix A identifies the acquisition within a staging. For edge A at staging 3, four acquisitions were performed, and the A1 designation was omitted from the first acquisition label. Therefore, R2-S3 is equivalent to R2-S3-A1. For the other three edges, two acquisitions were conducted at staging 3. These acquisitions are denoted by R2-S3 and R2-S3-A2. Within staging 3, pairwise CCFs between acquisitions were computed, and their mean and standard deviation were calculated. Variability between distinct stagings was assessed separately through the staging comparisons described above.

Each distinct unordered pair from the same edge was counted once, excluding self-comparisons. Replicate comparisons pair the scan of replicate 1 with every scan of replicate 2, including its repeated stagings and acquisitions. Staging comparisons pair scans of replicate 2 from different stagings, including all acquisitions within those stagings. Acquisition comparisons pair

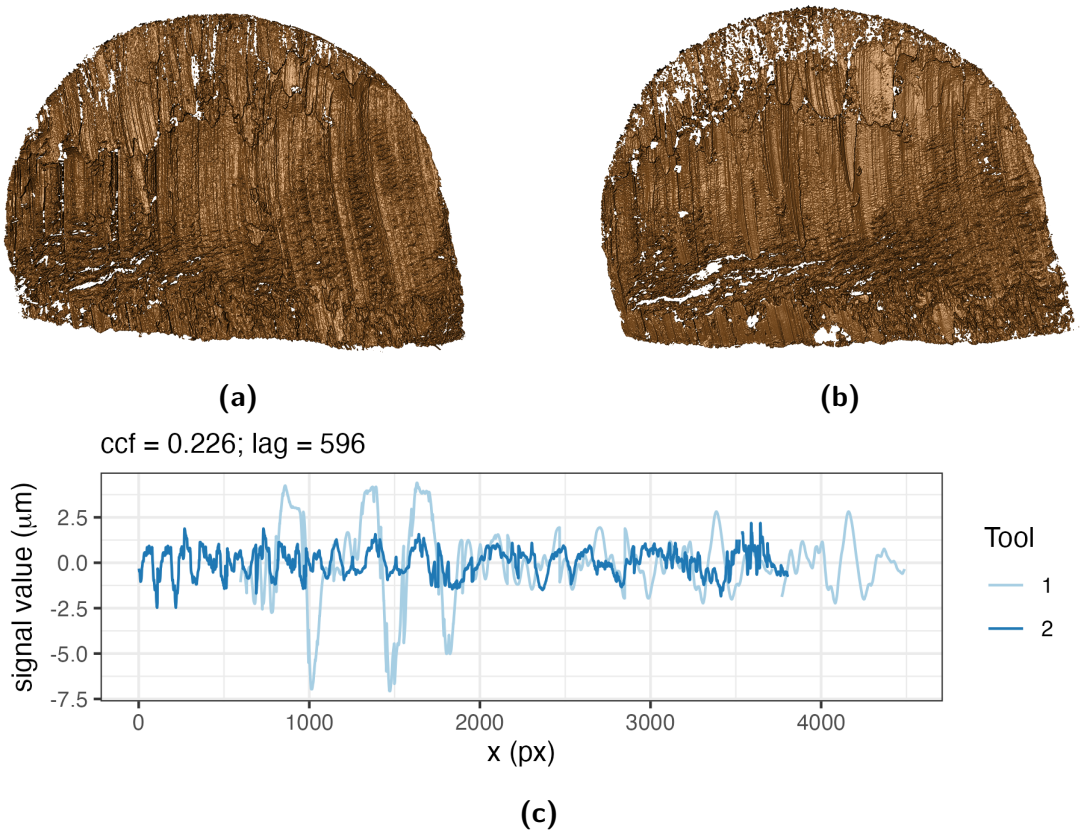


Figure 6. Different-source scan and signal comparison. (a) Scan T1AW-LI-R1 cut by tool 1. (b) Scan T2AW-LI-R1 cut by tool 2. (c) Alignment of signals from T1AW-LI-R1 and T2AW-LI-R1.

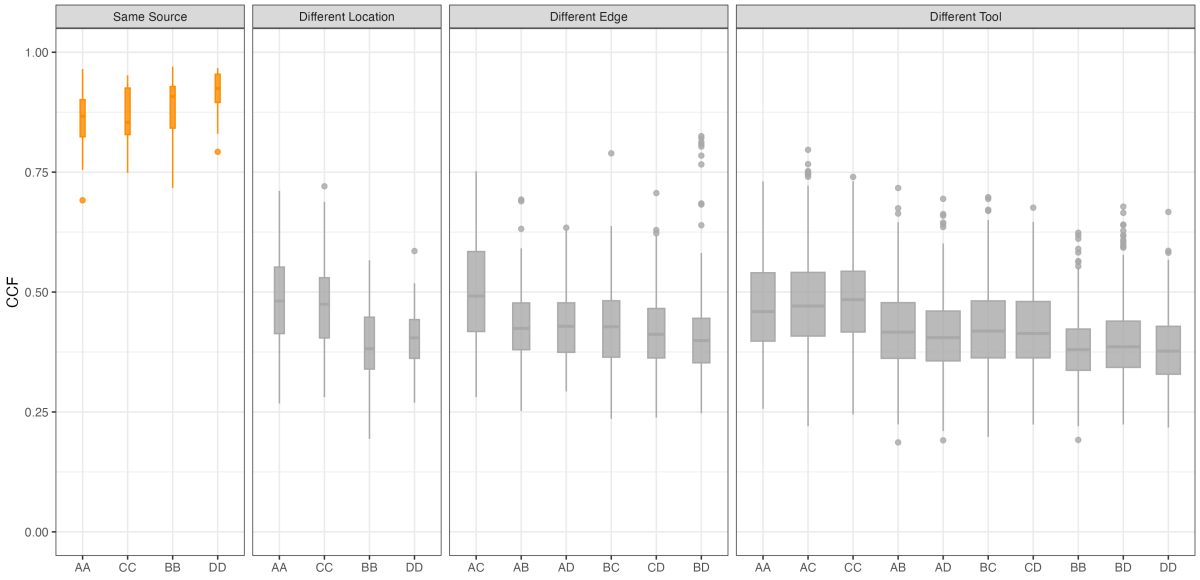


Figure 7. CCF distributions across comparison classes. Comparisons use the 120 main-experiment scans at staging 1 and acquisition 1. Facets are ordered from left to right by decreasing expected similarity: same source, different location, different edge, and different tool. Orange boxplots show same-source comparisons; gray boxplots show different-source comparisons. Horizontal-axis labels identify the paired edges.

scans from the same replicate and staging but different acquisitions. These categories contain,

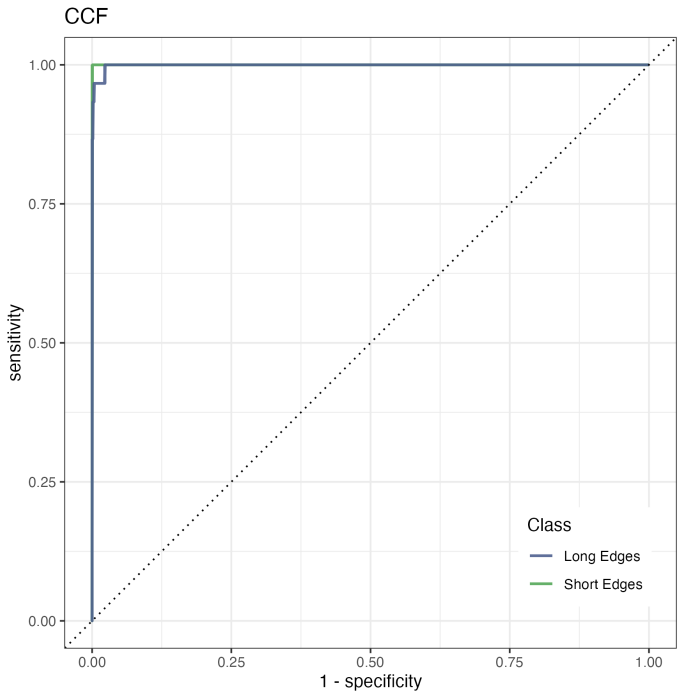


Figure 8. ROC curves for CCF-based classification. Comparisons use the 120 main-experiment scans at staging 1 and acquisition 1. Curves compare same-source pairs with different-tool pairs separately for long and short edge classes; same-tool different-source pairs are excluded. The dotted diagonal indicates chance-level classification.

respectively, 6, 9, and 6 comparisons for edge A and 4, 5, and 1 for each of edges B, C, and D. With this setup, CCF values were expected to become more consistent from replicate to staging to acquisition as the scanning conditions became progressively more stable. Consequently, variability was expected to decrease along this sequence. Replicates, stagings, and acquisitions were also compared, with smaller differences expected between acquisitions and stagings than between stagings and replicates. The results were visualized as scatter plots. Figure 9 shows the scatter plots and intervals extending two standard errors (SEs) above and below the mean. For each comparison category and edge, the plotted SE is calculated as $\mathrm{SD}/\sqrt{n}$, where n is the number of distinct pairwise CCF values. Higher orange mean lines indicate an increase in the mean CCF; interval widths depend on both the SD and the number of comparisons. Because comparisons share scans, these intervals are descriptive summaries rather than confidence intervals based on independent observations. The detailed numerical results, including means, standard deviations, and differences between means, are shown in Table 6. Mean CCF increases from replicate to staging to acquisition comparisons for every edge, and the acquisition-to-staging mean differences are smaller than the staging-to-replicate differences. The SDs do not decrease uniformly: edge B has a higher staging SD than replicate SD, whereas edges A, C, and D have lower staging SDs. For acquisition comparisons, SD is defined only for edge A, which has six pairs; the single pair for each other edge does not permit an estimate of dispersion.

[Agent: The manuscript contains 9 figures. Scientific Data recommends no more than eight but does not mandate this limit. Confirm with the coauthors or professor whether any logically connected pairs should be combined for revised submission; do not remove a figure or its discussion without an author decision.]

The variability assessment does not include variation in CCF values caused by differences in the manual extraction of profiles from scans. To support reproduction of the results reported here, the extraction locations and values for each scan have been included.

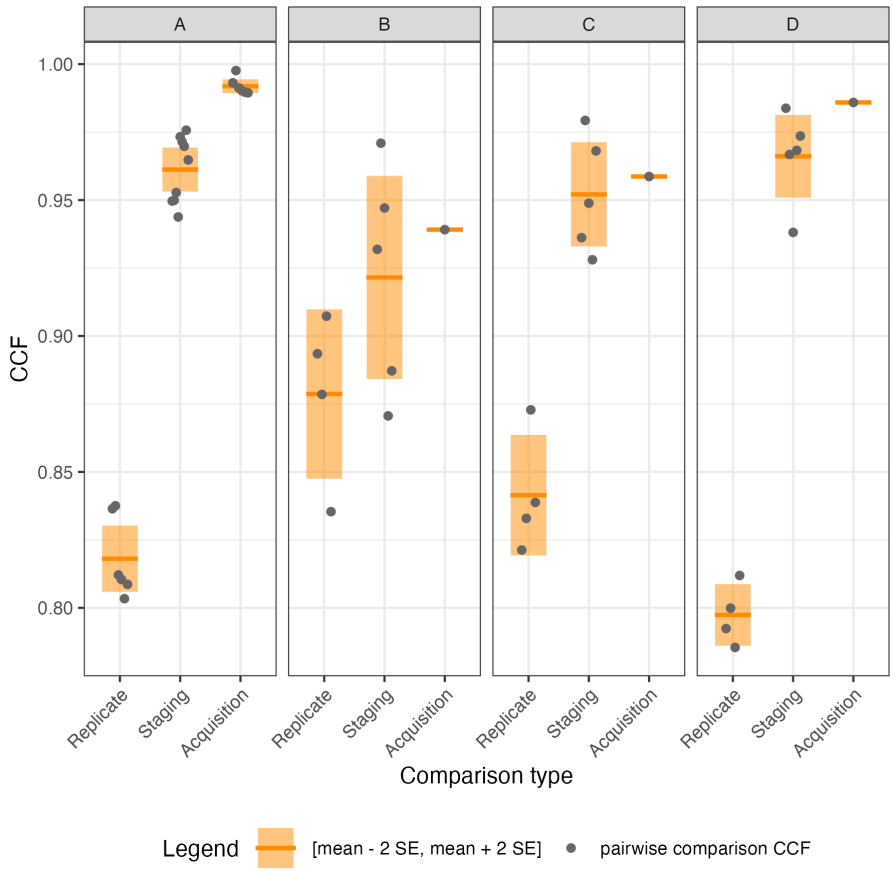


Figure 9. Variability across scanning conditions. Panels A, B, C, and D correspond to the four cut surfaces of tool 2 at the middle location, using 22 scans from the shared dataset. Gray points show pairwise CCF values, orange horizontal lines show means, and shaded intervals extend two standard errors above and below each mean, with SE calculated as SD divided by the square root of the number of comparisons. Each distinct unordered pair is counted once within the replicate, staging, or acquisition category; no interval is shown for a category with only one pair.

Table 6. Comparison of CCF means and standard deviations across different replicates, stagings, and acquisition settings. Dashes indicate that the standard deviation is undefined because only one CCF value is available.

	Replicate (R)		**Staging (S)**			**Acquisition (A)**		
Edge	**Avg$_R$**	**SD$_R$**	**Avg$_S$**	**SD$_S$**	**Avg$_S$ - Avg$_R$**	**Avg$_A$**	**SD$_A$**	**Avg$_A$ - Avg$_S$**
A	0.818	0.015	0.961	0.012	0.143	0.992	0.003	0.031
B	0.879	0.031	0.922	0.042	0.043	0.939	—	0.017
C	0.841	0.022	0.952	0.021	0.111	0.959	—	0.007
D	0.797	0.011	0.966	0.017	0.169	0.986	—	0.020

Usage Notes

The raw surface scans are provided in `x3p` format. To inspect or process these files in R, use the `x3ptools` package and its reference manual, available from CRAN[22]. The supplied profiles and processed signals provide a starting point for users who do not wish to repeat manual profile extraction. The supplied extraction coordinates allow these derived files to be related back to the corresponding scans. To reproduce the main-experiment summaries, select identifiers with

`staging = 1` and `acquisition = 1` in `meta.csv`, and retain pairwise rows only when both identifiers are in that set. For variability assessment, select tool 2 at the middle location and use the hierarchical comparison flags to distinguish replicate, staging, and acquisition comparisons as described in Technical Validation. Because each supplied signal is based on a manually selected profile, users assessing sensitivity to profile placement should re-extract profiles from the raw `x3p` scans using alternative documented locations.

Data Availability

The data and accompanying code described in this Data Descriptor are publicly available from Iowa State University DataShare[21] at `https://doi.org/10.25380/iastate.34025028`.

Code Availability

Custom R code was used to inspect raw scans, extract profiles, derive signals, align signals, and generate the validation outputs. The code is archived in the root-level `code/` folder of the Iowa State University DataShare deposit[21] at `https://doi.org/10.25380/iastate.34025028`. The files and their roles are described in Table 7.
The provided code implements the processing, comparison, validation, and visualization workflow described in Data Acquisition and Processing and Technical Validation. The supplied `renv.lock` records R 4.6.1, `x3ptools` 0.0.4, and the versions of the other R packages in the

Table 7. Overview of available code.

Code	Description	Section
Inspect raw scans		
`1-create_pngs_from_x3p.R`	obtain images of `x3p` files in `scans/`	Scan acquisition
Extract profiles		
`2-create_profiles_from_x3p.R`	manually extract profiles from each scan	Signal extraction
`3-create-single-profile-file.R`	combine profile information	Signal extraction
Derive signals		
`4-create_signals_from_profiles.R`	derive signals from each profile	Signal extraction
Compare signals		
`5-align-pairwise.R`	compute pairwise alignment CCF values	Pairwise alignment of signals
`6-all-comparison-results.R`	visualize main-experiment comparisons	Data Overview and Technical Validation
Assess variability		
`7-variability-assessment.R`	compare replicates, stagings, and acquisitions	Technical Validation

repository environment. For GitHub packages, it also records the source repositories and commit hashes. The archived `renv/` folder and `.Rprofile` project startup file support restoration of this environment.

References

[1] Lindley, D. V. A Problem in Forensic Science. *Biometrika* **64**, 207–213, 10.2307/2335686 (1977).

[2] AFTE. The association of firearm and tool mark examiners: Theory of identification as it relates to toolmarks. *AFTE Journal* **30**, 86–88 (1998).

[3] National Research Council (U.S.). *Strengthening Forensic Science in the United States: A Path Forward* (National Academies Press, 2009).

[4] President's Council of Advisors on Science and Technology. *Forensic Science in Criminal Courts: Ensuring Scientific Validity of Feature-Comparison Methods* (Executive Office of the President of the United States, President's Council, Washington, D.C., 2016).

[5] Biedermann, A. The strange persistence of (source) "identification" claims in forensic literature through descriptivism, diagnosticism and machinism. *Forensic Science International: Synergy* **4**, 15 pages, 10.1016/j.fsisyn.2022.100222 (2022).

[6] Bonfanti, M. S. & Ghauharali, R. I. Visualisation by confocal microscopy of traces on bullets and cartridge cases. *Science & Justice* **40**, 241–256, 10.1016/S1355-0306(00)71994-0 (2000).

[7] Smith, C. L. Linescan imaging of ballistics projectile markings for identification. In *Proceedings. 36th Annual 2002 International Carnahan Conference on Security Technology*, 216–222, 10.1109/CCST.2002.1049254 (2002).

[8] Li, D. Ballistics Projectile Image Analysis for Firearm Identification. *IEEE Transactions on Image Processing* **15**, 2857–2865, 10.1109/TIP.2006.877530 (2006).

[9] Chu, W. *et al.* Pilot Study of Automated Bullet Signature Identification Based on Topography Measurements and Correlations*†. *Journal of Forensic Sciences* **55**, 341–347, 10.1111/j.1556-4029.2009.01276.x (2010).

[10] McClarin, D. Adding an objective component to routine casework: Use of confocal microscopy for the analysis of 9mm caliber bullets. *AFTE Journal* **47**, 161–170 (2015).

[11] Baiker, M., Pieterman, R. & Zoon, P. Toolmark variability and quality depending on the fundamental parameters: Angle of attack, toolmark depth and substrate material. *Forensic Science International* **251**, 40–49, 10.1016/j.forsciint.2015.03.003 (2015).

[12] Garcia, D. L., Pieterman, R. & Baiker, M. Influence of the axial rotation angle on tool mark striations. *Forensic Science International* **279**, 203–218, 10.1016/j.forsciint.2017.08.021 (2017).

[13] Cuellar, M., Gao, S. & Hofmann, H. An algorithm for forensic toolmark comparisons. *Forensic Science International: Synergy* **9**, 100543, 10.1016/j.fsisyn.2024.100543 (2024).

[14] Ma, L. *et al.* NIST Bullet Signature Measurement System for RM (Reference Material) 8240 Standard Bullets. *Journal of Forensic Sciences* **49**, 1–11, 10.1520/JFS2003384 (2004).

[15] Zheng, X. A., Soons, J. A. & Thompson, R. M. NIST Ballistics Toolmark Research Database | NIST (2016).

[16] Chu, W., Thompson, R. M., Song, J. & Vorburger, T. V. Automatic identification of bullet signatures based on consecutive matching striae (CMS) criteria. *Forensic Science International* **231**, 137–141, 10.1016/j.forsciint.2013.04.025 (2013).

[17] Vorburger, T. *et al.* Applications of cross-correlation functions. *Wear* **271**, 529–533, 10.1016/j.wear.2010.03.030 (2011).

[18] Hare, E., Hofmann, H., Carriquiry, A. *et al.* Automatic matching of bullet land impressions. *The Annals of Applied Statistics* **11**, 2332–2356, 10.1214/17-AOAS1080 (2017).

[19] Ju, W. & Hofmann, H. An Open-Source Implementation of the CMPS Algorithm for Assessing Similarity of Bullets. *The R Journal* **14**, 267–285, 10.32614/RJ-2022-035 (2022).

[20] Cleveland, W. S., Grosse, E. & Shyu, W. M. Local Regression Models. In *Statistical Models in S* (Routledge, 1992).

[21] Lin, Y. *et al.* Dataset and code: Three-dimensional data of wire-cut surface scans under the confocal microscope, Iowa State University DataShare, https://doi.org/10.25380/iastate.34025028 (2026).

[22] Hofmann, H., Vanderplas, S., Krishnan, G. & Hare, E. "x3ptools: Tools for Working with 3D Surface Measurements", 10.32614/CRAN.package.x3ptools (2018).

Author Contributions

[Agent: Confirm whether all authors approve conversion of this statement to the CRediT taxonomy before revising the contributor roles.]
Y.L.: Methodology, Software, Validation, Data Curation, Writing: Draft; H.H.: Conceptualization, Methodology, Validation, Writing: Review and Editing; C.M.: Laboratory Supervision; E.A.: Physical Specimen, Scanning; J.S.: Forensic Advice; A.C.: Funding Acquisition.
All authors reviewed the manuscript.

Competing Interests

H.H. is a technical advisor to the Association of Firearm and Toolmark Examiners (AFTE), a fellow of the American Statistical Association (ASA), and a member of the ASA Forensic Science Committee. H.H. has testified as a court witness on behalf of Judge April Neubauer in the New York State Supreme Court, Criminal Term, in New York City. **[Agent: Confirm that the declaration covers every author and replace Alicia's declaration, if any.]**
The authors declare the following competing interests:

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